Carlo Alberto Giordano

Cybersecurity · Firmware & Hardware Reverse Engineering · Embedded Security

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◈ carloalbertogiordano.github.io/cv ⌂ Piedimonte Matese (CE), Italy

PROFILE

MSc in Computer Science (Cybersecurity) from the University of Salerno, graduated October 2026. I work security from the bottom up — firmware, embedded hardware and operating systems: reverse-engineered Intel ME firmware at ring -3, decrypted a car's firmware through a ZipCrypto weakness, built a checkm8-based recovery chain on my iPhone 5, torn down BLE and toner chips, and run a production Proxmox cluster. Looking for hands-on roles in penetration testing, reverse engineering and embedded/hardware security — hybrid or remote.

EXPERIENCE

Software Engineer — R&D Collaboration
Jul 2026 – present

Flytech S.r.l. — drone inspection & PV thermography · Remote

Built an internal tool that streamlines the identification of damaged photovoltaic panels from drone imagery, outperforming DJI Terra on the company's own dataset. Proprietary — not published.

IT Consultant & Systems Management
2018 – present

Giordano Moda — Piedimonte Matese (CE)

- Developed and maintained in-house management software and automations in Python and Bash.
- Integrated AI workflows into business processes through custom-built software.
- Designed and run a production Proxmox cluster backing the company infrastructure.
- Database management; staff training and direct client-facing support.

Production-flow Analysis
Summers 2022 – 2025

Specialcorde srl — Gioia Sannitica (CE)

Mapped and analysed production processes, identified bottlenecks and proposed corrective actions, presenting findings directly to management.

SECURITY PROJECTS

Intel ME (CSME 11.8) Security Assessment
2026

Authorized university offensive-security project on a Surface Pro 4. Reverse-engineered the Intel Management Engine FTPR modules (BUP, MCA) in Ghidra at ring -3, applied the full penetration-testing methodology, verified the firmware's patch state against known advisories (INTEL-SA-00086, CVE-2019-0090) and identified several original, previously-undisclosed findings beyond public literature. Reconstructed and statically validated the firmware-to-JTAG exploitation chain. *Details held under responsible disclosure.*

`Intel ME / CSME` `Ghidra` `Firmware RE` `ring -3`

iPhone 5 (A6) Passcode Recovery — Security Research
2026

End-to-end bootrom-exploit recovery chain on a personally-owned, locked iPhone 5 (A6, no Secure Enclave): checkm8 DFU exploit → unsigned SSH ramdisk → system keybag extraction → a custom on-device brute-forcer driving the hardware AES engine, plus the kernel patch needed to unlock it. Reached the same per-attempt speed as commercial forensic tooling; documented as a rigorous negative result.

`checkm8` `iOS` `Firmware RE` `Forensics` ↗ github

TP-Link TL-WR841N — Firmware RE & Custom Firmware
2024

with Mattia (@Cvlmtt)

Reverse-engineered the router firmware (binwalk, Firmware-Mod-Kit, firmwalker), built a MIPS32R2 / uClibc-ng cross-toolchain with Buildroot, and discovered the undocumented layout constraints (SquashFS block size + exact padding) required to rebuild a working custom firmware that preserves configuration persistence.

`Firmware RE` `MIPS` `Buildroot` `Embedded`

HP Toner Chip — Hardware Reverse Engineering 2026

Hardware-level pentest of an HP cartridge "E-label" chip (chip-on-board). Pinout mapping with a multimeter in diode mode, OneWire/raw-probe and current-modulation analysis via Arduino — with positive/negative controls that caught three false positives (including 50 Hz mains hum). A rigorous negative result on the chip's protocol, plus HP Dynamic Security context.

Hardware RE · Electronics · Arduino · COB

Monitor Firmware RE (Realtek RTD2525BE / 8051) 2026

Dumped a monitor's SPI flash (CH341A), reverse-engineered the 8051 scaler firmware in Ghidra and decoded its bank-switching to chase a start-up brightness bug. Root-caused it to a backlight apply-chain never run at boot; two hardware patch attempts documented honestly, including a self-caught no-op test.

8051 · Ghidra · SPI dump · Firmware RE

XY Tag (RM1230) BLE Tracker Teardown 2026

Hardware and firmware teardown of a Chinese AirTag-clone BLE tracker: PCB analysis, debug-pad identification, firmware dump and static analysis in Ghidra.

BLE · Hardware RE · Ghidra · ↗ github

Mazda Firmware Decryption 2024

Reverse-engineered a Mazda infotainment firmware update: broke its ZipCrypto encryption via a known-plaintext attack (bkcrack) and documented the full decryption procedure.

Firmware RE · Cryptanalysis · ZipCrypto · ↗ github

Binary Exploitation & Mitigation (Nebula / Protostar) 2025

Exploitation and defensive write-ups for the exploit.education training VMs: each finding mapped to a CWE with an attack tree, exploited (up to a remote stack-overflow → root), and then fixed with corrected, compilable C.

Binary Exploitation · CWE · Secure Coding

Fixedness — Data Privacy Auditing Framework (MSc Thesis) 2026

Framework that measures the risk of logically reconstructing sensitive records in anonymised datasets, via reduction to SAT/SMT and detection of deterministic leaks (co-NP UNSAT). Evaluated across 10 anonymisation techniques. Developed with Prof. Ugo Fiore (UNISA). Paper in preparation — repository private.

Python · SAT/SMT · Data Privacy · Research

OS Development 2023 – 2024

Bare-metal kernels in C, x86 Assembly and Rust — bootloader, memory management, interrupt handling — built to understand attack surfaces at the hardware level.

C · Assembly x86 · Rust · ↗ myOS · nonameOS · MyRustyOS

TECHNICAL SKILLS

Reverse Engineering / Offensive

STRONG Firmware RE · Ghidra · Hardware RE

GOOD Binary exploitation · HID/BadUSB · Metasploit

GOOD Hashcat · John the Ripper · bkcrack · Volatility

FAIR Cryptanalysis · Wireshark · tcpdump

Languages

STRONG C · Python · Bash · Java

FAIR Assembly (x86) · Rust

Infrastructure

STRONG Linux · Proxmox VE · PBS · TrueNAS

EDUCATION

MSc — Computer Science (Cybersecurity curriculum) 2023 – 2026
University of Salerno · LM-18

BSc — Computer Science 2020 – 2023
University of Salerno

High School Diploma — Applied Sciences 2016 – 2020
Liceo Scientifico G. Galilei — Piedimonte Matese (CE)

LANGUAGES · CERTIFICATIONS · INTERESTS

LANGUAGES

Italian — Native
English — C1 (Advanced)

CERTIFICATIONS

English C1 Certificate
Obtained in Ireland

INTERESTS

Embedded & firmware security
Homelab · Motorcycle track days

I authorise the processing of my personal data in this CV pursuant to Art. 13 GDPR (EU 2016/679) and Italian Legislative Decree 196/2003.